EECE 571L Deep Learning in Digital Media
Course Project Report

Age-Invariant Face Recognition

Yiming(Eric) Cao
Yanjun Meng
Akhil Rajavaram

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University of British Columbia
Department of Electrical and Computer Engineering

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1 method

In this section, we provide an overview of the proposed method to verify facial images across different ages, the proposed OursKD model framework, the loss function that applies in the OursKD model, the datasets used in the experiments.

This section outlines the architecture of the OursKD model, depicted in the figure. Initially, a facial image serves as input, which is then processed through both the teacher and student backbones to generate feature maps. Next, the student's feature map is inputted into the student model's head, producing the student's logits—a vector of probabilities that determine the class to which the input facial image belongs.

1.1 Loss function

In OursKD model, we hope to transfer knowledge from teacher to student to get a better model that can gain advantage of both models. So we connect teacher with student on the loss function, which leads to student can learn some knowledge from teacher for each times the neural network updates its weight. Therefore, we have below configurations:

For facial image $x \in X$, it has one-hot label y, and we feed this image x to the teacher's backbones can get its feature maps $F_t(x)$, and feed this image x to the student's backbones can get its feature maps $F_s(x)$.

If we regard the process of transfer knowledge is a process of teaching, then input x is the question teacher want to teach, and the feature map $F(x)$ is the answer of question, so we want student's answer as close as teacher's answer. So we need to make $F_t(x)$ as close as $F_s(x)$, then we need to apply L_2 loss on $F_t(x)$ as close as $F_s(x)$, which is mean squared error between teacher and student's backbone's output feature map.

On the other hand, the teacher's feature map $F_t(x)$ cannot 100% represent the facial image x, so to prevent overfitting of the student's model from teacher's model, we feed the student's backbone's output feature map to the student's classification head and get the probabilities vector $H_s(F_s(x_i))$ and apply cross-entropy on image x's ground-truth label y with that.

$$\mathrm{L} = \mathrm{L2}(F_t(x), F_s(x)) + \mathrm{CE}(y', y) \tag{1}$$

1.2 Preformance metrics

To evaluate the OursKD model, we cannot just measure the student's model's prediction accuracy, that can be easily calculate by counting the percentage of student model predicted correctly, we also want to measure how well student's backbone's output feature map can represent the input image. So we need to combine these two terms together, so we proposed a way for our own model.

We used two different facial images x_i and x_j as input and a label $z_{i,j}$ that indicates if they belong to the same person. If $z_{i,j} = 0$, then it belongs to different facial class; if $z_{i,j} = 1$, then it belongs to the same facial class with different ages.

So we measure the mean squared error between the two feature maps $F_s(x_i)$ and $F_s(x_j)$, and selcet a threshold that idcates two feature maps belong to same identity. We cannot just set threshold to be zero, since the two facial image with same identity but different age will produce a similar but not same feature maps. Therefore, we select the best threshold that has largest accuracy.

We assume that, if the mean squared error between the two feature maps match to the best threshold we choose, then these two images are from same facial class with different ages. If not, they are belong to total different people.

With the proposed method, OursKD models are comprised of two submodels: the student model and the teacher model. To transfer knowledge from the teacher model to the student model, the teacher model requires pretrained weights containing essential "knowledge". Additionally, loading pretrained weights into the student model can further enhance the performance of OursKD. Therefore, we pretrained the ElasticFace and MTLFace as the submodels.

As mentioned above, we choose ElasticFace and MTLFace as our submodels and results are shown in the table. The main reasons for these choices are the following unique properties of each network.

References